Project Title: “Development of a Web-Based Integrated School Management System for Xiamarah Wisdom School Inc.”

RESEARCHER

Agocoy, Michael John L. (BSIT3-09)
Ormilla, Apple Jane H. (BSIT3-08)
Parcia, Chemie V. (BSIT3-09)
Sambaan, Elija (BSIT3-10)
Zagado, Vincent Jade A. (BSIT3-05)

SEPTEMBER 2026

I. Title of the Study

Development of a Web-Based Integrated School Management System for Xiamarah Wisdom School Inc.

(The title reflects the main purpose of the study, which is to develop a web-based system that integrates and manages the different processes of Xiamarah Wisdom School Inc., including student records, enrollment, class and schedule management, attendance, grades, payments, announcements, and report generation.)

II. Introduction

The use of technology in education has become increasingly important in improving the management of school operations and services. Schools handle various activities such as student enrollment, student records, class schedules, attendance, grades, payments, and communication, which require organized and accessible information. With the growing use of digital technologies, centralized systems can help schools manage information more efficiently and reduce the need for separate records and manual processes. Calibuso (2024) emphasized the importance of web-based school management systems in addressing problems associated with manual record management, including information mismanagement, slow retrieval, lack of organization, and limited accessibility. In the case of Xiamarah Wisdom School Inc., many of these processes are still handled through manual methods and Microsoft Excel.

However, managing different school processes through separate or manual methods can cause difficulties in organizing, updating, and retrieving information. School staff may spend more time recording data, checking records, and preparing reports, while information may become duplicated or difficult to track. Existing methods may also have limited coordination between enrollment, student records, attendance, grades, payments, and other school activities. Calibuso (2024) found that manual record management can result in difficulties in organizing, retrieving, and managing school information, highlighting the need for a web-based management system to improve the accessibility and efficiency of school records. These concerns show the need for a system that can connect these processes and provide a more organized way of managing school information.

To address these concerns, this study proposes the Development of a Web-Based Integrated School Management System for Xiamarah Wisdom School Inc. The proposed system will integrate important school processes into one web-based platform, including student management, enrollment, class and schedule management, attendance, grade management, payment records, online class management, announcements, notifications, and report generation. The system aims to improve the efficiency, accessibility, accuracy, and organization of school information. This study will benefit the school administration and staff by making school processes easier to manage, teachers by providing convenient access to relevant student and class

information, students by providing more organized school services, and parents or guardians by providing more organized school-related information and updates.

III. Statement of the Problem

The current school management processes of Xiamarah Wisdom School Inc. face several limitations:

1. **Manual enrollment process** – Enrollment still involves manually completed forms and processing, which can take more than 15 minutes depending on the time needed to complete the forms.
2. **Separate payment records** – School payments are recorded using Excel, and transferring information from official receipts to Excel is identified as a major problem.
3. **Difficulty in managing student information** – Student records and class information are handled through manual records and different systems, making information more difficult to organize, update, and access.
4. **Lack of centralized school management** – The school does not have one system that connects enrollment, student records, payments, class schedules, and online classes. The school expressed a need for one system to manage these processes more efficiently.
5. **Limited automation and reporting** – Several school activities still require manual recording and checking, creating a need for automatic records, easier information identification, payment tracking, and report generation.

Thus, there is a need for a web-based integrated school management system that can centralize school information, reduce manual processes, automate important tasks, and improve the efficiency of managing school operations at Xiamarah Wisdom School Inc.

IV. Objectives of the Study

General Objective:

To develop and implement a web-based integrated school management system that will centralize and improve the management of school processes at Xiamarah Wisdom School Inc.

Specific Objectives:

1. To gather and analyze data on the current school processes, including enrollment, student records, payments, class schedules, online classes, communication, and report preparation.
2. To design a web-based integrated school management system that will organize and connect the different school processes in one centralized platform.
3. To develop system modules for student management, enrollment, payment management, class and schedule management, attendance, grade management, online class management, announcements, notifications, and report generation.
4. To implement automated features that can reduce manual recording and make student information, payment records, and school reports easier to manage and access.
5. To evaluate the system's functionality, usability, and effectiveness through testing and feedback from the school's administrators, teachers, and staff.

V. Significance of the Study

This project will benefit the following stakeholders:

Students – Benefit from more organized enrollment, student records, class schedules, payments, and online class information.

Teachers – Have easier access to relevant student information, class schedules, online class management, and other school-related records.

School Administration and Staff – Improve the management of enrollment, student records, payments, schedules, and reports through one centralized system.

Parents/Guardians – Receive more organized school information, announcements, payment records, and other updates related to their students.

Future Researchers – Use the system as a reference for developing further improvements and related school management systems.

VI. Scope and Delimitation

Scope:

- Development of a web-based integrated school management system for Xiamarah Wisdom School Inc.
- Core features: student enrollment, student records management, payment management, class and schedule management, attendance, grade management, online class management, announcements, notifications, and report generation.
- The system will provide automated and centralized management of school information to reduce manual recording and make records easier to access and manage.
- User roles: Administrator, Teacher, and Student.

Delimitation:

- The system will focus only on the school processes and requirements identified during the study and will not include features outside the school's identified needs.
- The system will not replace external systems such as the DepEd Learner Information System (LIS); it will only manage and organize the school's internal records.
- Online classes will focus on managing class information and related activities within the system and will not replace third-party platforms such as Google Meet.
- The system will be developed and tested specifically for Xiamarah Wisdom School Inc. during the study.

VII. Methodology

The study will adopt a Developmental Research approach using the Agile methodology for iterative design and testing.

Development Tools:

Frontend: ReactJS / Angular

Backend: Node.js with Express

Database: MySQL / PostgreSQL

Proctoring Tools: JavaScript-based browser activity monitoring, OpenCV for webcam snapshots

Analytics: Data visualization with Chart.js or D3.js

Research Process:

1. Requirements Gathering – Interviews with faculty and IT staff.
2. System Design – Use case diagrams, database schema, and system architecture.
3. Development – Incremental feature implementation using Agile sprints.
4. Testing – Unit testing, security testing, and user acceptance testing.
5. Evaluation – Surveys and feedback from pilot users to measure usability and effectiveness.

VIII. Review of Related Literature

Grepon et al. (2022) discussed the development of an e-School System for a community college in Northern Mindanao, Philippines. The system was developed to centralize school and student data and support different school transactions. The study included functions related to admission, enrollment, accounting, student information, grading, and report generation. The researchers found that the system was functional, usable, and reliable. This study is closely related to the proposed Web-Based Integrated School Management System because both aim to centralize school processes and reduce the workload associated with manual record management.

Vidanagamage et al. (2022) developed a web-based Student Management System using Node.js, React.js, and MongoDB. The system was designed to improve the effectiveness and efficiency of teaching, learning, and administrative activities through organized student information management. The study demonstrates how web technologies can be used to provide an authenticated and centralized platform for educational institutions. This supports the proposed system's student management and web-based architecture.

Nugroho, Mandyartha, and Munir (2022) examined the implementation of a Student Management System at TK Al-Wahyu Surabaya. The study identified difficulties caused by manually managing student information through paper records and computer files, including time-consuming data processing and risks of human error. The proposed system helped organize student data and reporting. This is relevant to the present study because Xiamarah Wisdom School also needs a centralized system to reduce manual recording and improve access to student information.

The OECD (2023) explained that digital education systems can support institutional management through tools for student information, attendance, classes, grades, enrollment, admissions, reporting, and communication. The report emphasizes that integrating digital tools can support administrative and educational operations. This literature directly supports the proposed system because its major functions include student records, enrollment, class schedules, attendance, grades, notifications, and reports.

Roman, Roman, and Fiscal (2023) examined digital maturity in a basic education institution and its implications for school management and operations. Their study considered digital practices used in school management and examined their effects on institutional operations. The findings showed the importance of digital practices in supporting school management. This supports the present study's objective of introducing a web-based system to improve the management and organization of school processes.

Wahyudin, Wahyudi, and Komarudin (2023) developed a web-based school academic information system for an MTs school in Bandung. The existing processes involved manually managing student, teacher, subject, schedule, and class information, which created risks of data

loss and inefficiency. The developed system helped curriculum officers and teachers manage academic information more efficiently. This study is relevant to the proposed system's student management, class scheduling, and academic record functions.

Octaviany, Rahman, and Mudjisusatyo (2024) developed and evaluated a web-based Academic Management Information System. Their study found that integrating technology into academic management could reduce administrative burdens, save time, improve resource utilization, and support decision-making. The system was evaluated by experts and users and was considered suitable for managing academic activities. This supports the proposed system's goal of automating school processes and improving administrative efficiency.

Alabin et al. (2024) presented the School Management System–BCPAMS, a web-based academic management system for Bestlink College of the Philippines. The system focused on scheduling, grade management, student records, faculty reports, and report monitoring. Its purpose was to make academic transactions easier and more efficient for administrators and other school personnel. This is related to the proposed system because it also includes student records, schedules, grades, and reporting functions.

Firmansyah, Setiyawan, and Turmudi (2024) developed a web-based school information system to improve administrative and communication efficiency. The study reported improvements in managing student data, grades, and attendance, as well as communication between schools, students, and parents. The researchers also identified staff training and infrastructure as important considerations when implementing the system. This supports the proposed system's student information, attendance, grades, announcements, and notification features.

Wahyuni, Muthmainnah, Permana, and Novalima (2025) reviewed the development of web-based School Management Information Systems and found that these systems can improve the efficiency and effectiveness of educational administration. Their review identified applications involving student data, teacher data, financial information, and reporting. The findings support the use of an integrated web-based platform to manage different school processes in one system. This is highly relevant to the proposed system because it combines student records, enrollment, payments, academic activities, and reports in a centralized platform

IX. Expected Output

The final output will be a functional web-based integrated school management system with the following deliverables:

- Centralized school management portal for authorized users.
- Administrator features for managing student records, enrollment, payments, classes, schedules, attendance, grades, announcements, notifications, and reports.
- Teacher features for accessing relevant student information, managing class-related records, attendance, grades, and online class information.
- Student features for accessing enrollment-related information, class schedules, attendance, grades, payment information, announcements, and online class information.
- Automated and organized reports for school records and management needs.

The system is expected to reduce manual recording, improve access to school information, support more organized school processes, and make school records and reports easier to manage.

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XI Appendices

Appendix A – Consolidated Data

The interview questionnaire was conducted with **Elgen L. Yamit, School Administrator of Xiamarah Wisdom School Inc.**, to understand the school's current processes and problems.

Interview Questionnaire: Xiamarah Wisdom School Inc.

Respondent Name: Elgen L Yamit

Position/Office: School Administrator

Date of Interview: Sep 12, 2026

I. Current Enrollment

Questions	Response
1. How is the enrollment process currently done in the school?	- Manual process like fillup in a form.
2. How are student enrollment records recorded and stored?	- Recorded in a paper only and stored in the envelope.
3. How long does it usually take to enroll one student?	- It takes 15 minutes.
4. Who is responsible for processing and checking enrollment records?	- Admin and Supervisor.
5. What problems do you usually experience during the manual enrollment process?	- We sometimes experience errors when filling out the enrollment forms, such as missing or incorrect information.
6. How do you correct or update a student's enrollment information?	- We check the form and manually correct the wrong information.

II. School Payment Management

Questions	Response
1. What school payments do students need to pay aside from enrollment fees?	- Enrollment Fee, books, and miscellaneous fee
2. What payment methods does the school accept for tuition and other school fees?	- Through cash and Gcash.
3. How do you record and monitor student payments?	- They record and monitor payments using excel.
4. How do you know if a student has fully paid, partially paid, or still has a balance?	- They track the remaining balances, fully paid, partially paid using excel.
5. How do you handle late payments or unpaid balances?	- They monitor late payments and unpaid balances using Excel.
6. What problems do you experience when recording and checking student payments?	- They sometimes experience errors, duplicate entries, and mistakes in recording student payments, which can make it difficult to accurately track payment records and balances.

III. Student Records Management

Questions	Response
1. What student information does the school currently keep?	- The school keeps basic student information such as name, age, address, contact information, grade level, and other enrollment-related details.
2. How are student records organized and updated?	- Student records are organized and updated manually.
3. How do you search for a student's record when needed?	- They search for the student's record manually through folders.
4. How do you keep track of active, pending, or inactive students?	- They monitor the students' status manually through their records.
5. Who is allowed to access student records?	- Only the teachers and administrator.
6. What problems do you experience when managing student records manually?	- They experience difficulty in searching and updating records, possible errors in recording information, and duplicate or misplaced records.

IV. Online Class Management (During Pandemic)

Questions	Response
1. What platform or application is used for online classes?	- They only used messenger apps.
2. How are students informed about their online class schedules?	- Using messenger group chats.
3. How do teachers check student attendance during online classes?	- They only manually record using attendance recorded.
4. How do students submit their assignments or activities?	- During face to face class.
5. What problems do teachers and students experience during online classes?	- They experience internet connection problems, difficulty communicating, and challenges in accessing online class materials.

V. Class and Schedule Management

Questions	Response
1. How are class schedules currently created?	- Class schedules are created manually based on the available teachers, subjects, classrooms, and students.
2. How are students informed when there are changes in the class schedule?	- Students are informed through messages, announcements, or directly by the teacher
3. How do you assign students to their classes or sections?	- Students are assigned to classes or sections based on their grade level, subjects, and available class slots.
4. How do teachers receive their class schedules?	- Teachers receive their schedules through printed copies, messages, or direct communication from the school.
5. How do you handle schedule conflicts or changes?	- Schedule conflicts or changes are checked manually, and adjustments are made by coordinating with the teachers and students.

VI. Teacher and Staff Management

Questions	Response
1. What information about teachers needs to be recorded?	- Teacher's name, information, educational background, teaching experience, and subjects they can teach.
2. How are teachers assigned to subjects, classes, or sections?	- Teachers are assigned based on their subjects and available schedules.
3. How do teachers access the student information they need for their classes?	- Teachers access student information through school records.
4. What features would make the work of teachers and staff easier?	- Easy access to student records, schedules, attendance, and announcements.

VII. Communication and Notifications

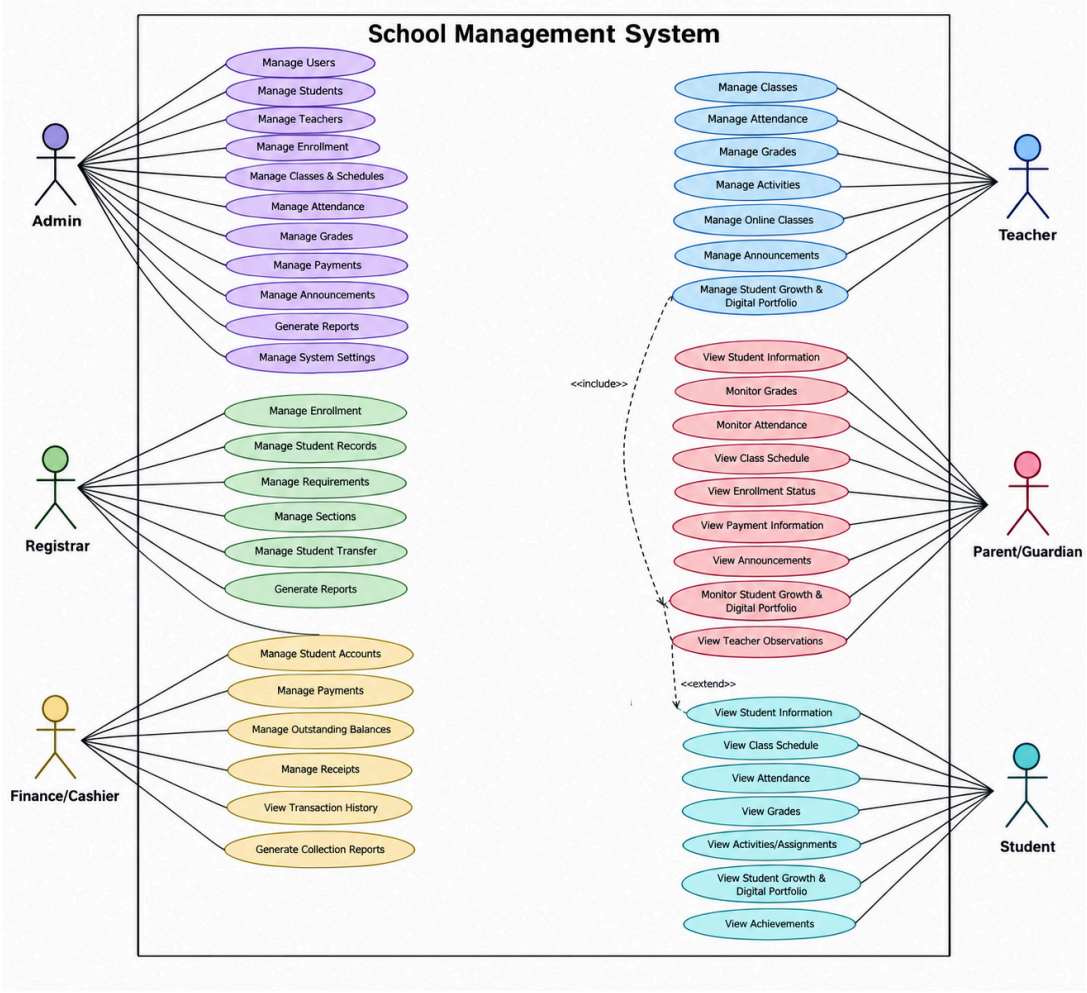
Questions	Response
1. How does the school communicate important announcements to students and parents?	- Through messages, announcements, or social media.
2. How are students informed about enrollment schedules and requirements?	- Through announcements, messages, or social media posts.
3. How are students informed about changes in class schedules?	- Through messages or announcements from the school or teachers.
4. How are reminders for classes, assignments, or school activities given?	- Through messages or announcements from the teachers.
5. What communication problems does the school currently experience?	- Sometimes students may not receive announcements or reminders on time.

VIII. Reports and Monitoring

Questions	Response
1. What reports does the school currently prepare about enrollment?	- The school prepares enrollment reports showing the number of enrolled students.
2. How do you monitor the total number of enrolled students?	- By checking the enrollment records and counting the enrolled students.
3. How do you keep track of students by grade level or section?	- By organizing the student records according to their grade level and section.

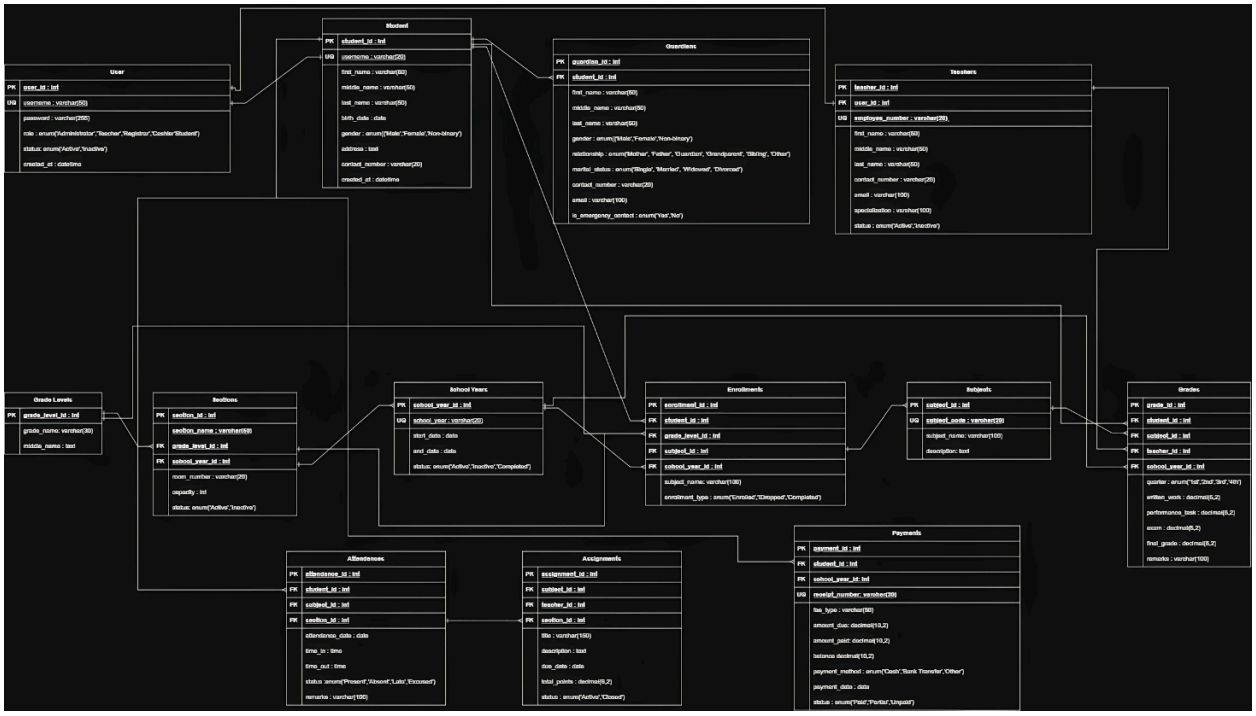
Appendix B – Use Case Diagram

Diagram showing interactions among Admin, Faculty, and Students.



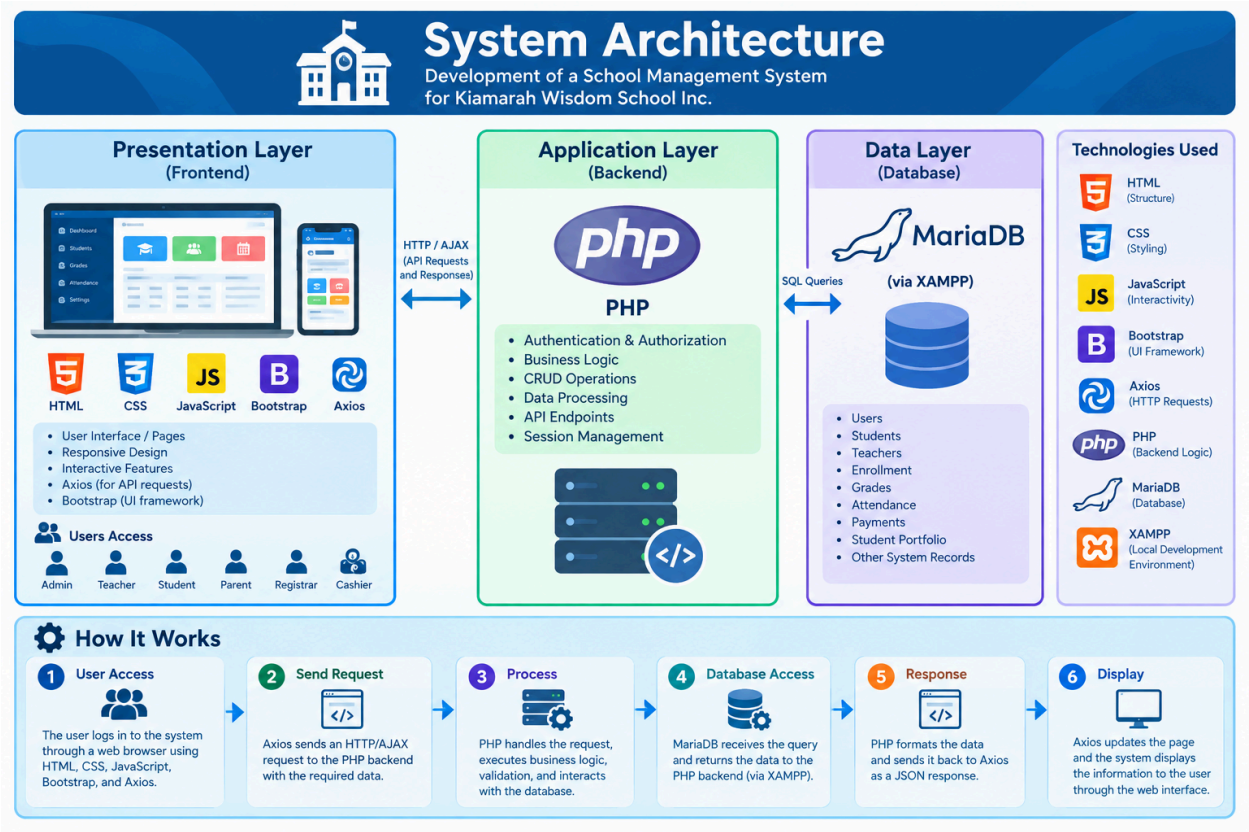
The use case diagram shows how each user role interacts with the School Management System. It shows the specific features available to the Admin, Registrar, Teacher, Cashier, and Parent/Guardian, ensuring that each user can only access functions related to their role.

Appendix C – Entity-Relationship Diagram (ERD)



The ERD illustrates how student, guardian, teacher, enrollment, academic, attendance, assignment, and payment data are interconnected within the school management system.

Appendix D – System Architecture

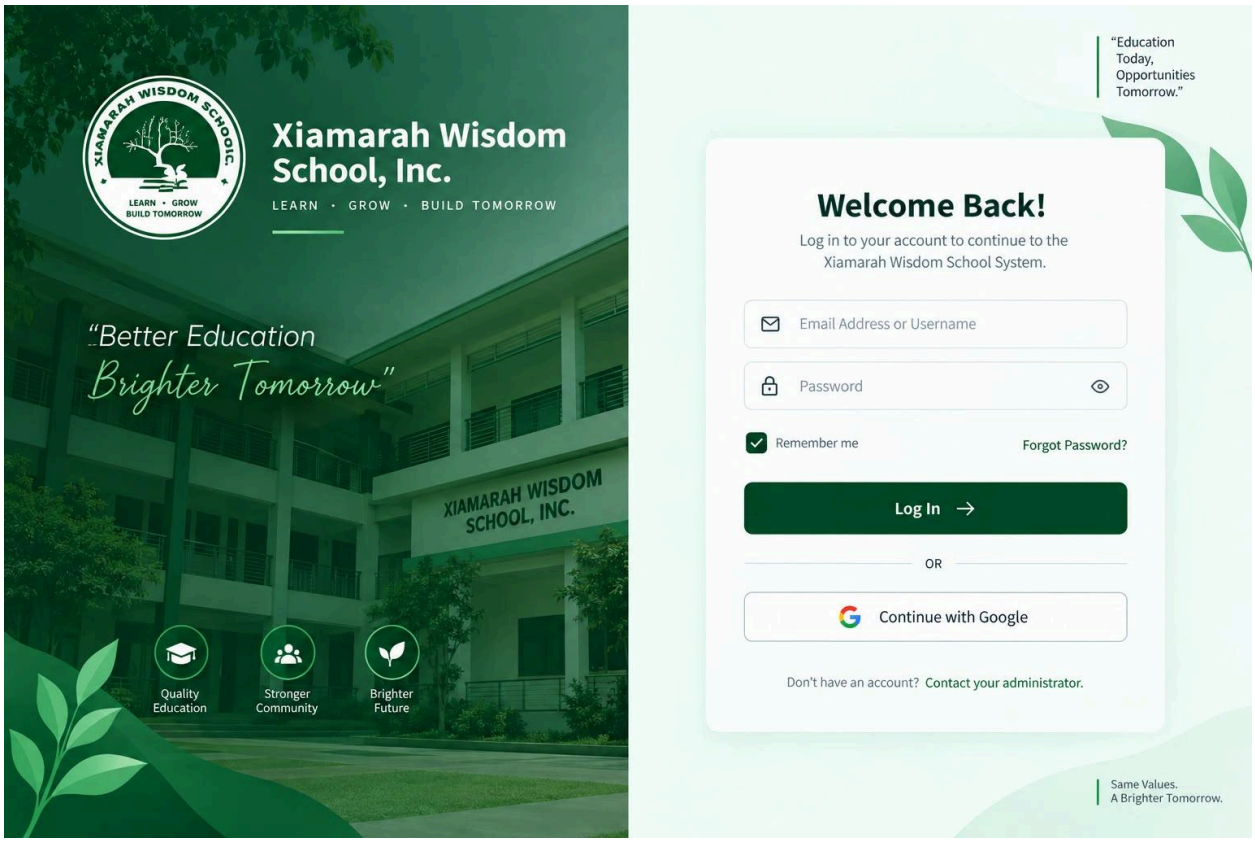


A 3-tier diagram (Frontend – ReactJS, Backend – Node.js/Express, Database – MySQL).

The proctoring module is integrated into the system through the Presentation Layer, where students access the online assessment and proctoring interface. The frontend sends proctoring-related data and assessment requests to the Application Layer through API requests. The backend processes the assessment and proctoring information and communicates with the Data Layer to store relevant records such as examination results, session information, and proctoring logs. This allows the proctoring module to work together with the existing student, assessment, and user management functions of the school management system.

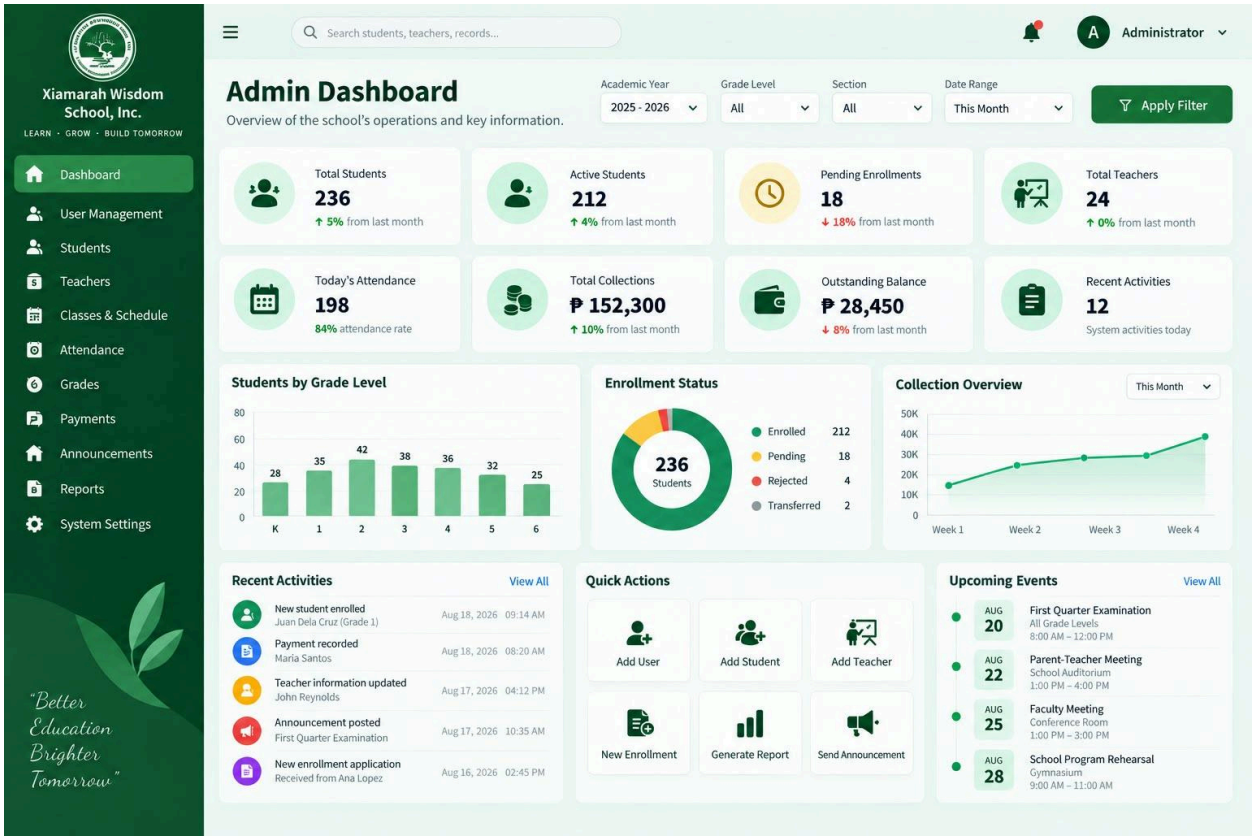
Appendix E – Figma UI/UX Prototypes

ADMIN DASHBOARD



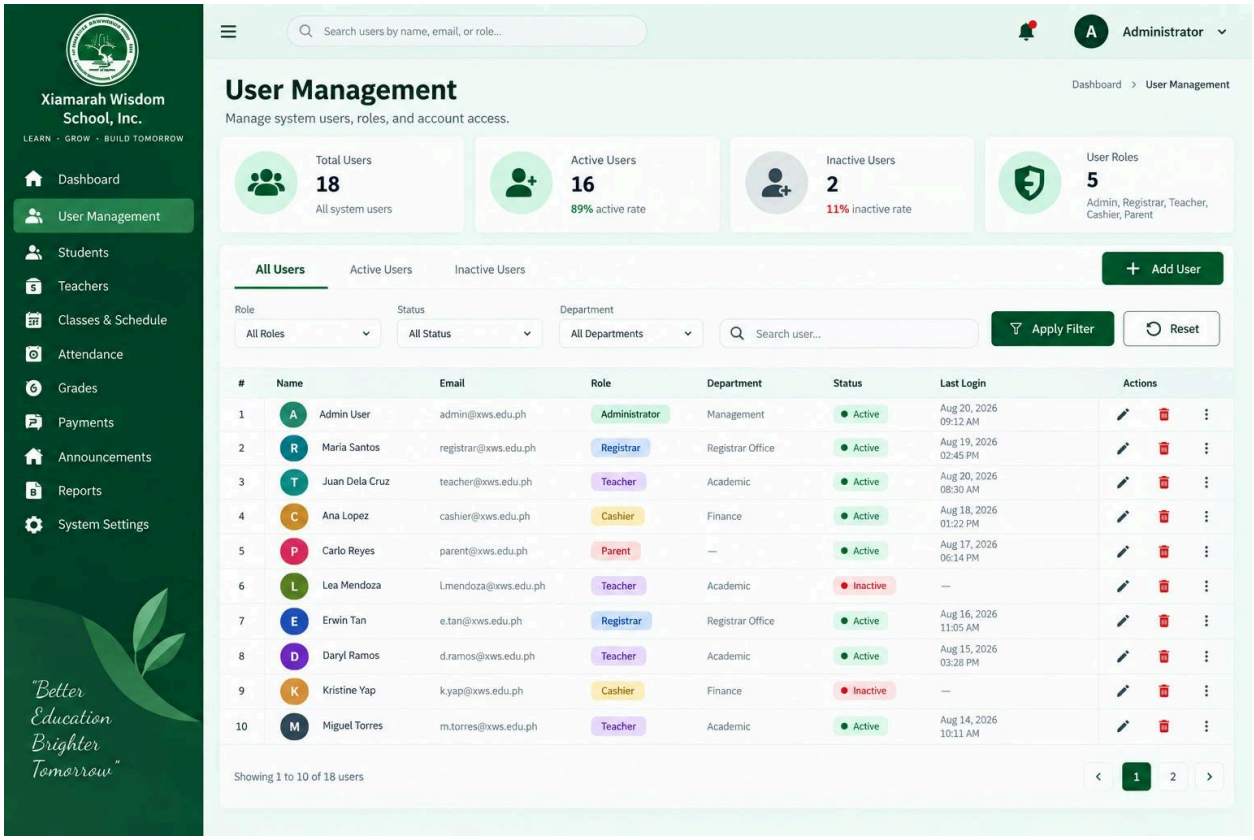
1. Login Page

The Login page allows authorized users to securely access the school management system using their username/email and password. It also provides password recovery and Google sign-in options.



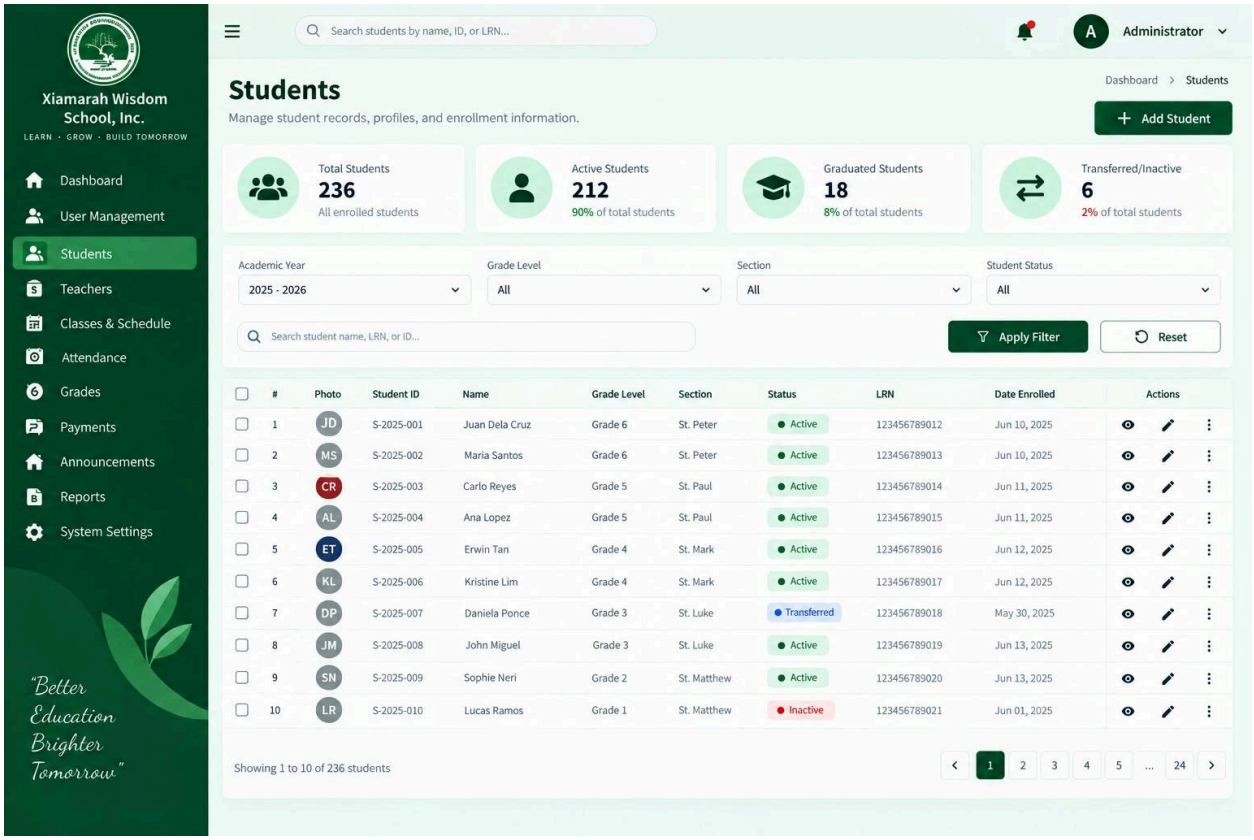
2. Admin Dashboard

The Admin Dashboard provides an overview of school operations, including student enrollment, attendance, teachers, collections, balances, recent activities, and upcoming events.



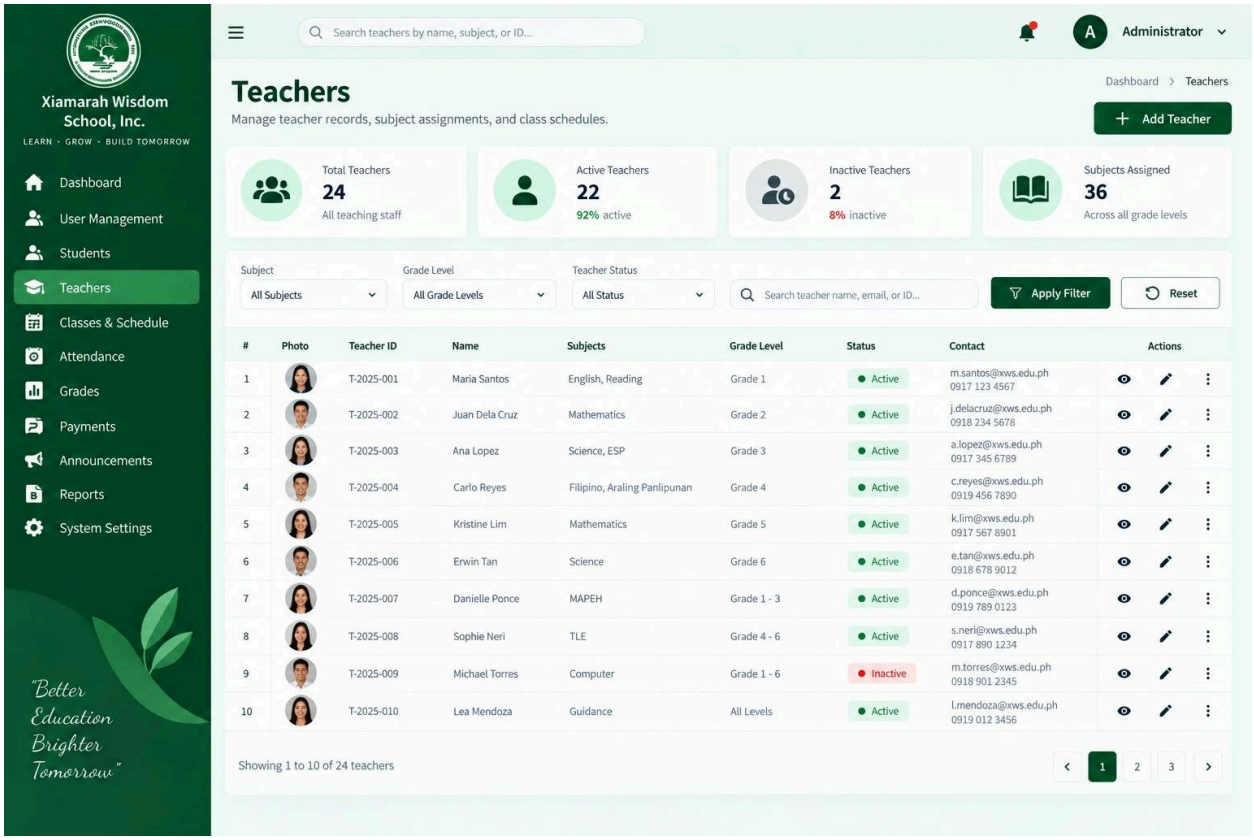
3. User Management

The User Management page allows administrators to manage system accounts, assign user roles, and control account status. It includes search, filtering, editing, and account management actions.



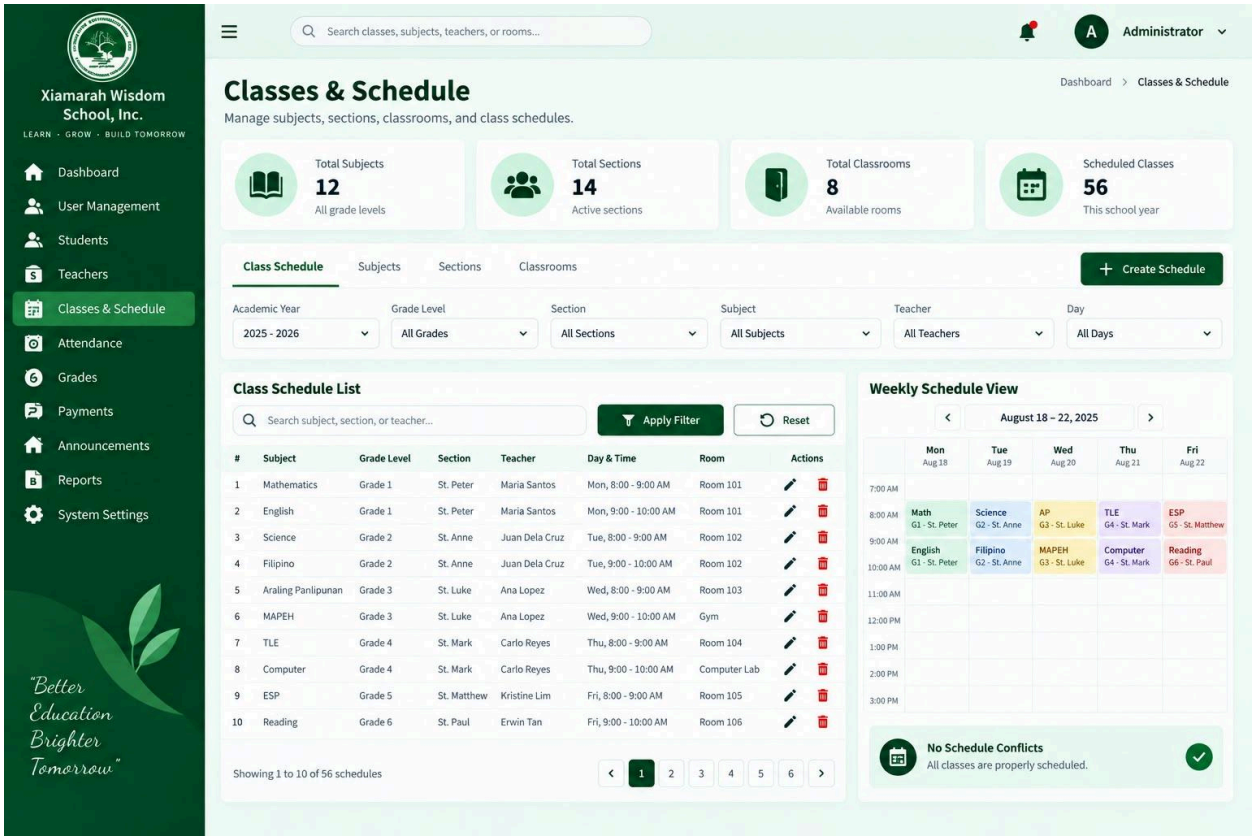
4. Students

The Students page allows administrators to manage student records, profiles, enrollment information, and student status. Search, filters, and pagination make it easier to locate specific students.



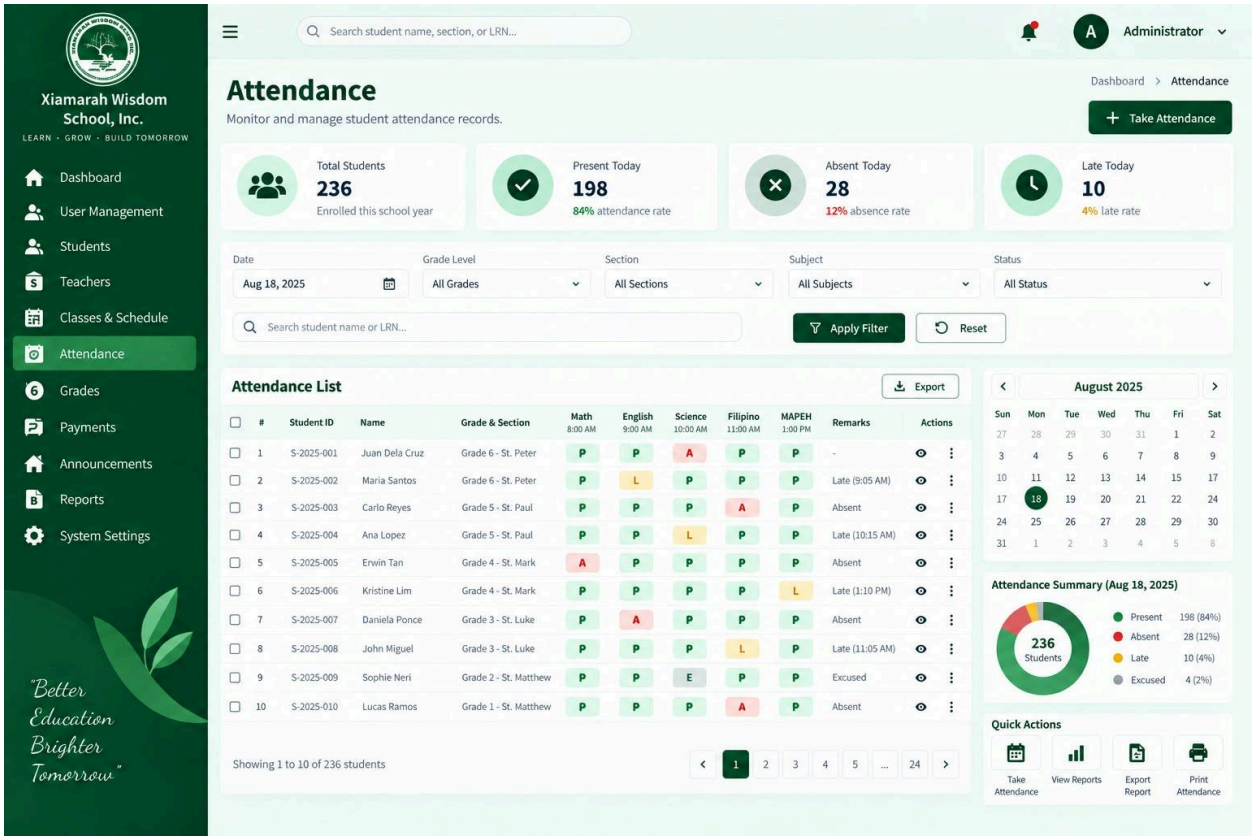
5. Teachers

The Teachers page allows administrators to manage teacher information, subject assignments, grade levels, and teacher status. It also provides search, filtering, and record management actions.



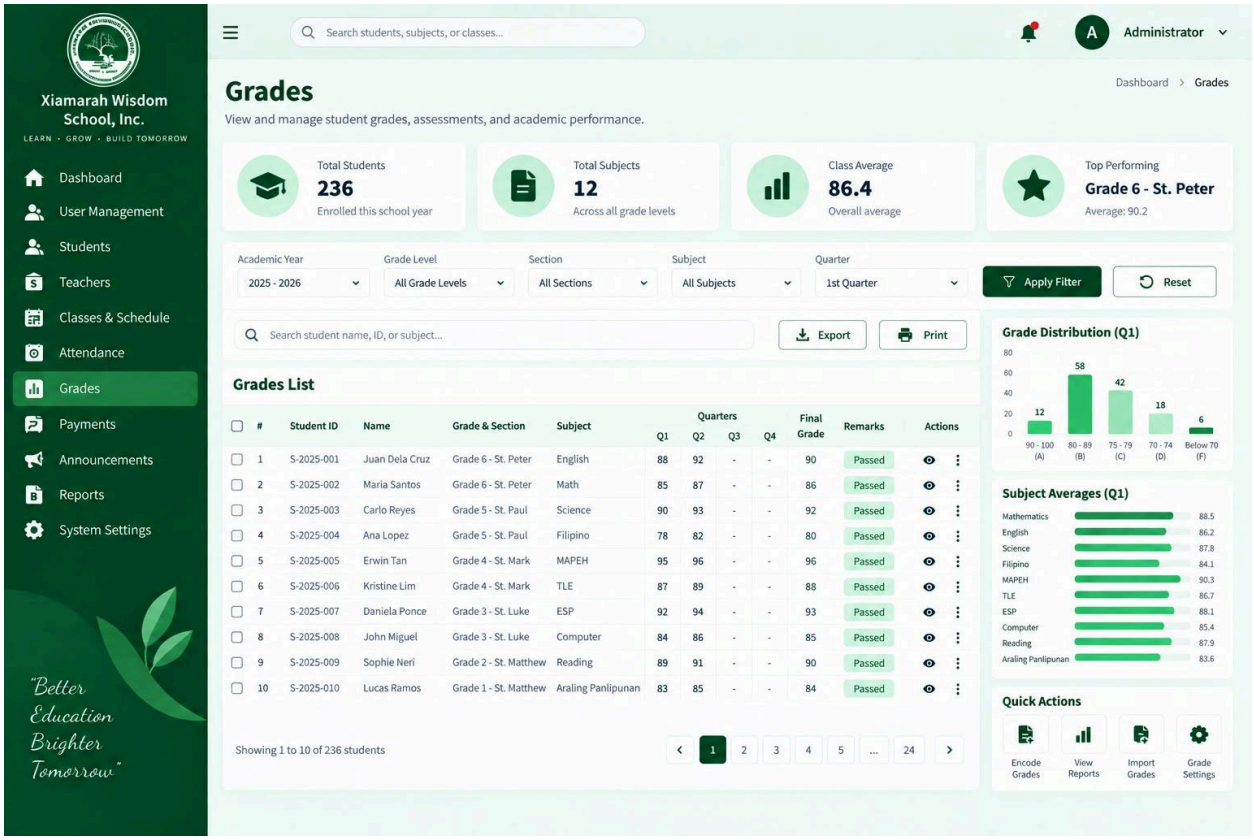
6. Classes & Schedule

The Classes & Schedule page manages subjects, sections, classrooms, teachers, and class schedules. The weekly schedule view helps users monitor class times and identify scheduling conflicts.



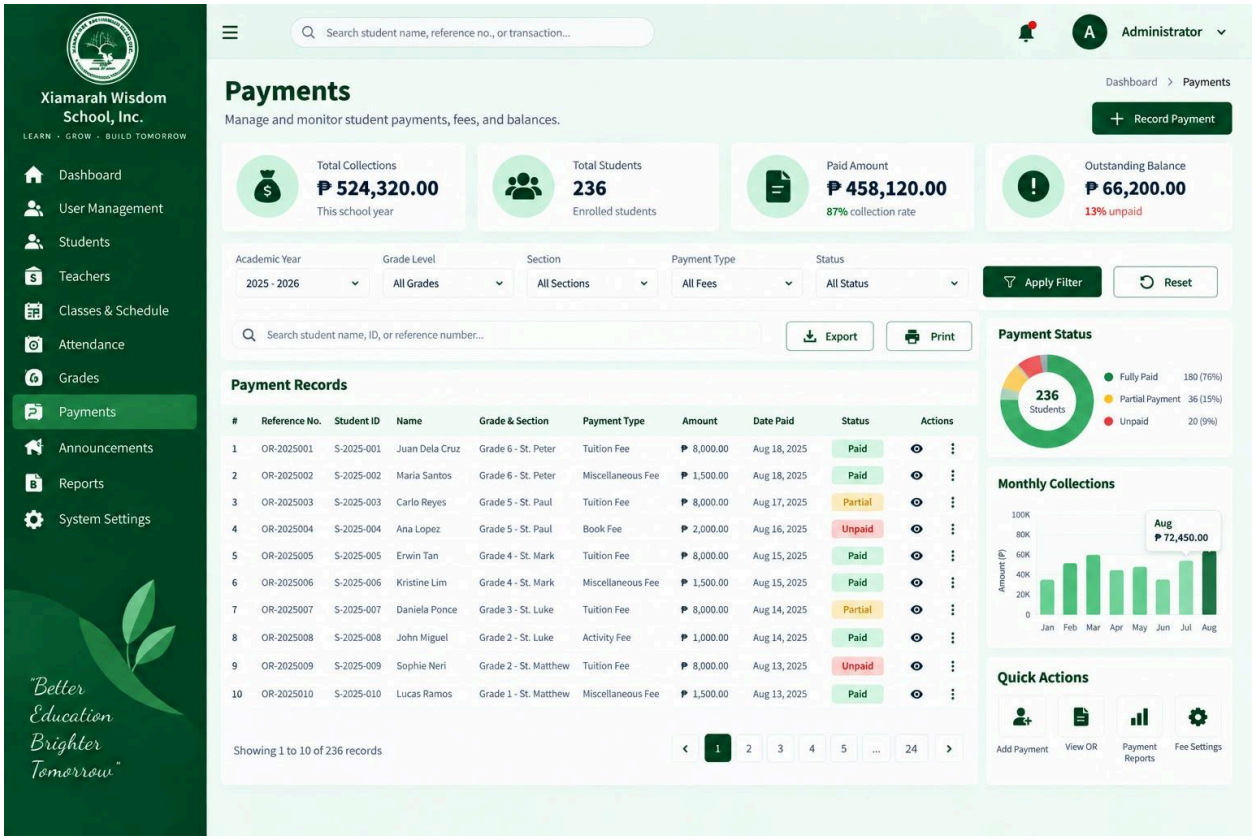
7. Attendance

The Attendance page allows authorized users to record and monitor student attendance. Attendance can be marked as P (Present), A (Absent), or L (Late), with filters and attendance summaries.



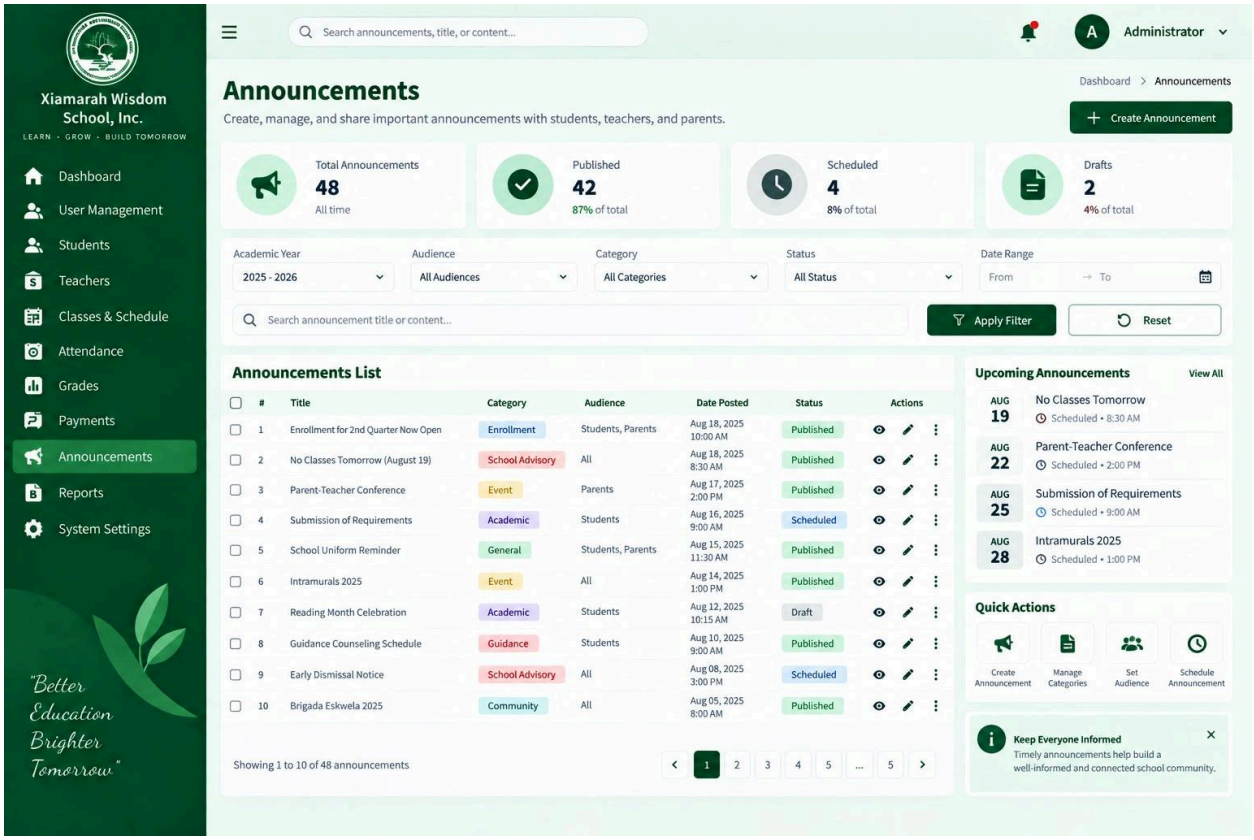
8. Grades

The Grades page allows teachers and authorized users to record, view, and manage student grades by subject and grading period. It also provides grade distribution and subject performance summaries.



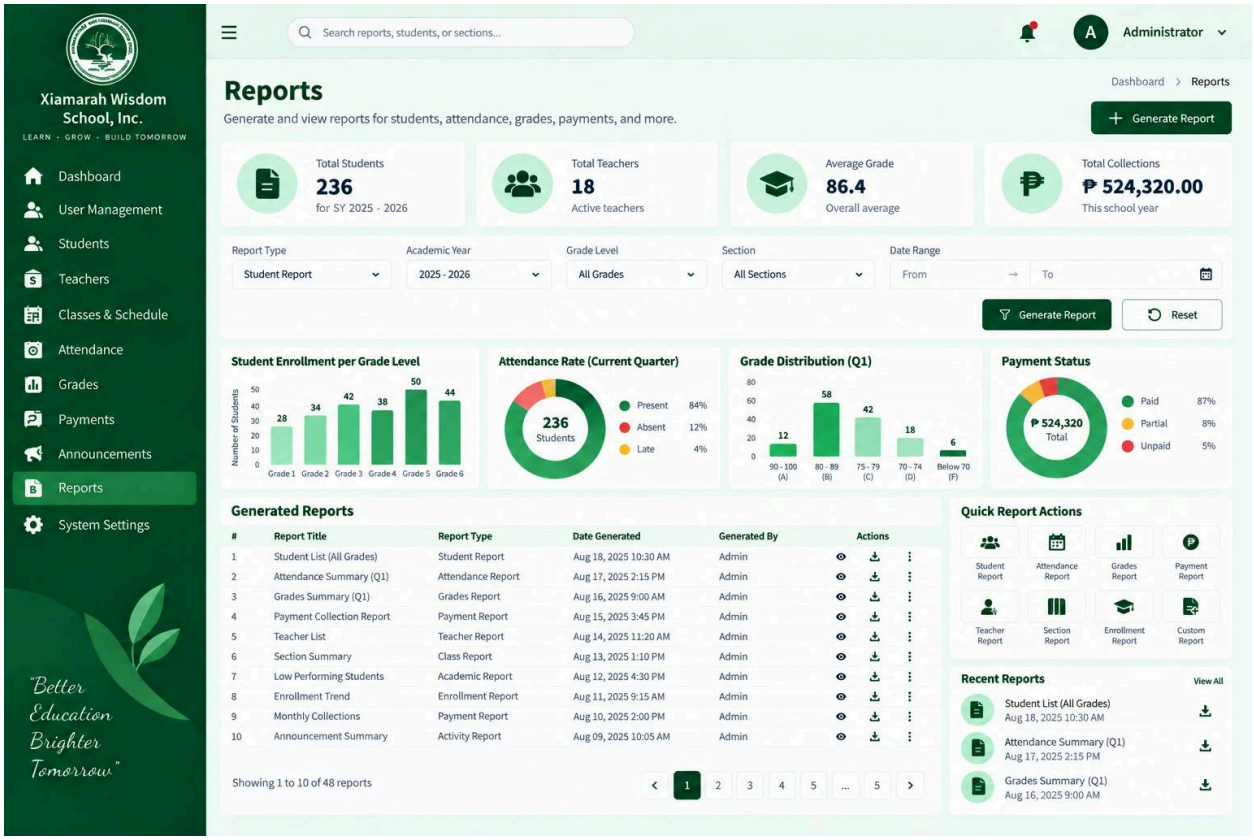
9. Payments

The Payments page allows authorized users to record and monitor student payments, school fees, and outstanding balances. Payment records can be searched, filtered, viewed, and organized by status.



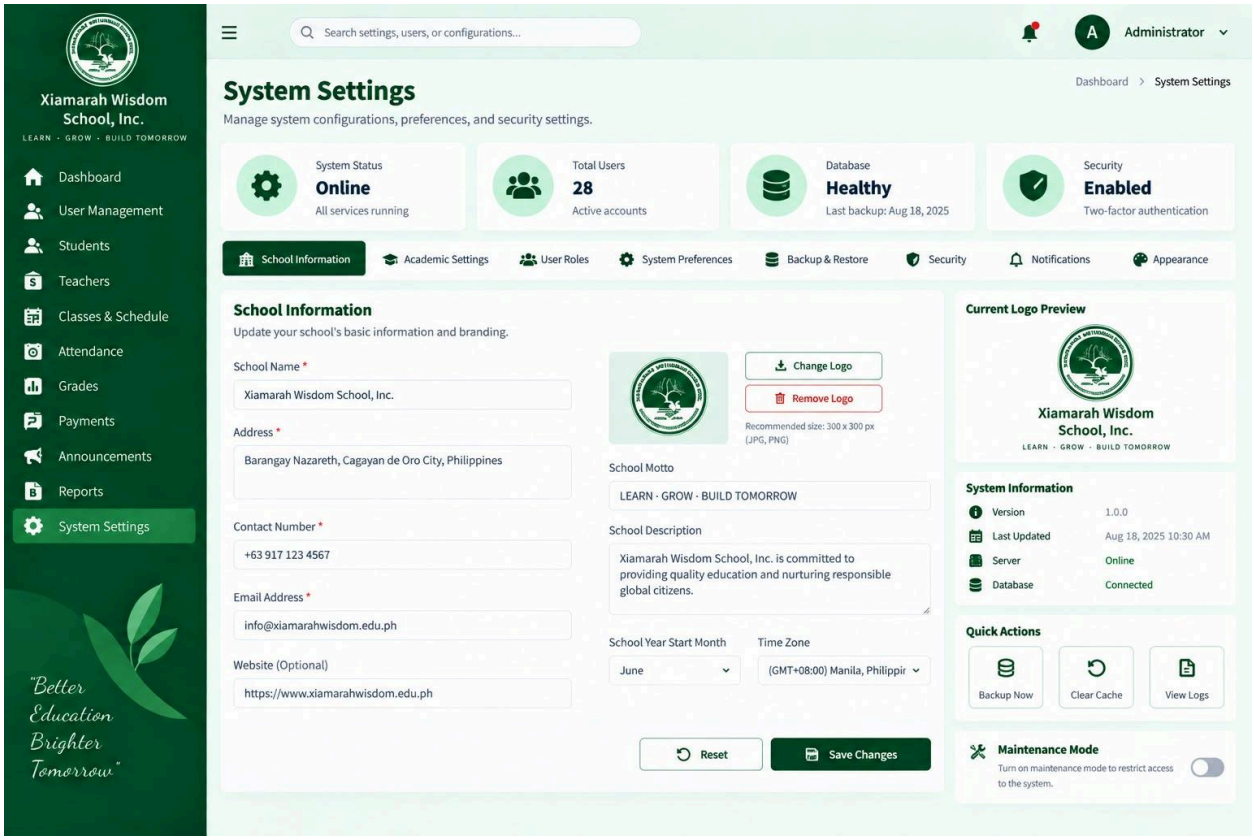
10. Announcements

The Announcements page allows administrators to create, manage, schedule, and publish important school announcements for students, parents, and teachers.



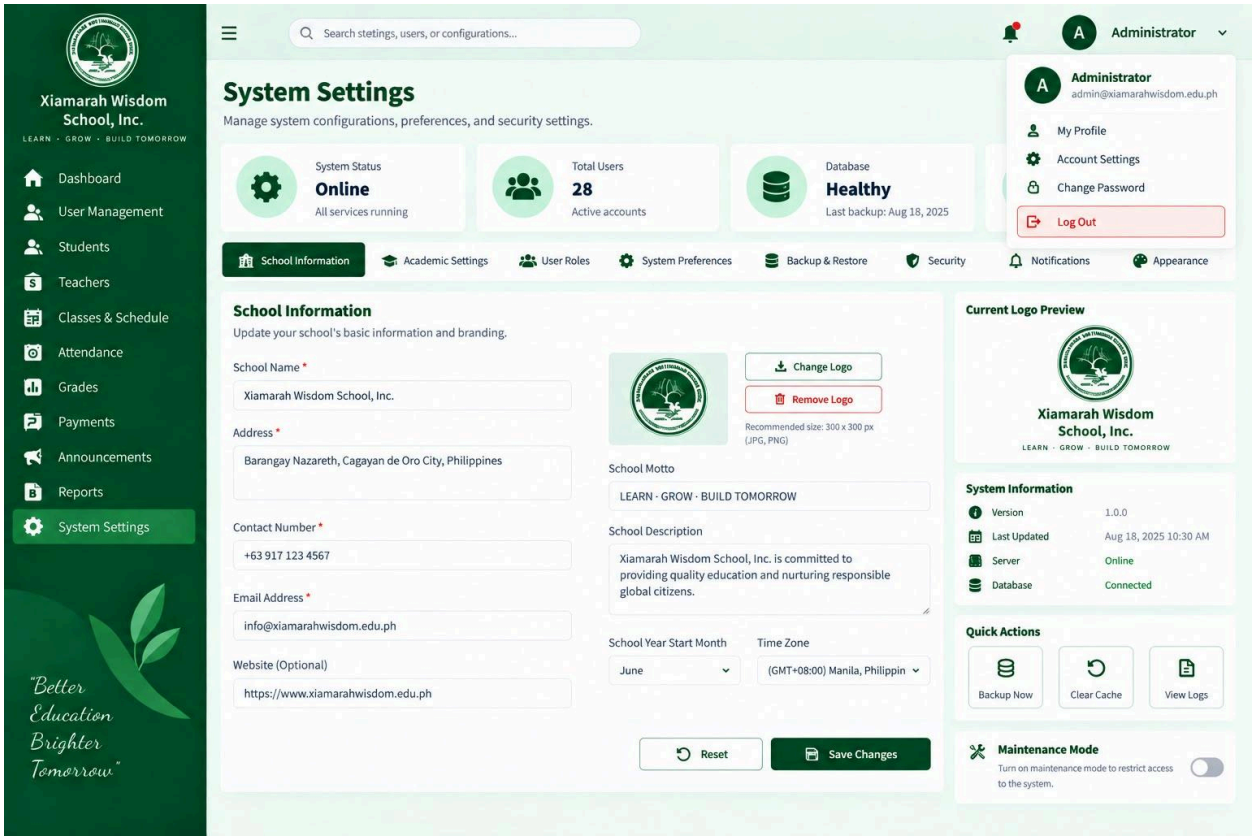
11. Reports

The Reports page allows administrators to generate, view, and manage different school reports, including student, attendance, grades, payment, teacher, and enrollment reports.



12. System Settings

The System Settings page allows administrators to configure school information, academic settings, user roles, security, notifications, system preferences, and backup options.



13. Administrator Profile / Log Out

The Administrator profile menu provides access to My Profile, Account Settings, and Change Password. The Log Out option allows the administrator to securely end the current session.

Appendix F – Testing Instruments (for later phases)

System Usability Scale (SUS) Questionnaire (English)

Name (optional): _____ **User ID:** _____ **Age:** _____

Sex: ☐ Male ☐ Female

Questions	Stronger Disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
	1	**2**	**3**	**4**	**5**
1. I think that I would like to use this system frequently.					
2. I found the system unnecessarily complex.					
3. I thought the system was easy to use.					
4. I think that I would need the support of a technical person to be able to play this system					
5. I found the various functions in this system were well integrated.					
6. I thought there was too much inconsistency in this system.					
7. I would imagine that most people would learn to use this system very quickly.					
8. I found the system very cumbersome to use.					
9. I felt very confident using the system.					
10. I needed to learn a lot of things before I could get going					

with this system.					
Total Score:					
Overall Score:					

System Usability Scale (SUS) Questionnaire (Bisaya)

Pangalan (kapilian): _____ **User ID:** _____ **Edad:** _____

Kasarian: ☐ Lalake ☐ Babae

Mga Pangutana	Kusog nga dili mouyon	Dili Mouyon	Wala ko kabalo sa akong opinyon	Uyon	Uyon Kaayo
	1	**2**	**3**	**4**	**5**
1. Sa akong hunahuna gusto nako nga gamiton kini nga system kanunay.					
2. Akong nakita nga ang system dili kinahanglan nga komplikado.					
3. Naghunahuna ko nga ang system sayon gamiton.					
4. Sa akong hunahuna kinahanglan nako ang suporta sa usa ka teknikal nga tawo aron makagamit niini nga system.					
5. Akong nakaplagan nga ang lain-laing mga gimbuhaton niini nga system maayo ang pagkahiusa.					
6. Naghunahuna ko nga adunay daghan kaayo nga					

pagkasumpaki niini nga system.					
7. Akong mahanduraw nga kadaghanan sa mga tawo makakat-on sa paggamit niini nga system nga dali kaayo.					
8. Akong nakita nga ang system lisud kaayo gamiton.					
9. Mibati ko og dako nga pagsalig sa paggamit sa system.					
10. Kinahanglan kong makat-on og daghang mga butang sa dili pa ako makapadayon niini nga system.					
Total Score:					
Overall Score:					